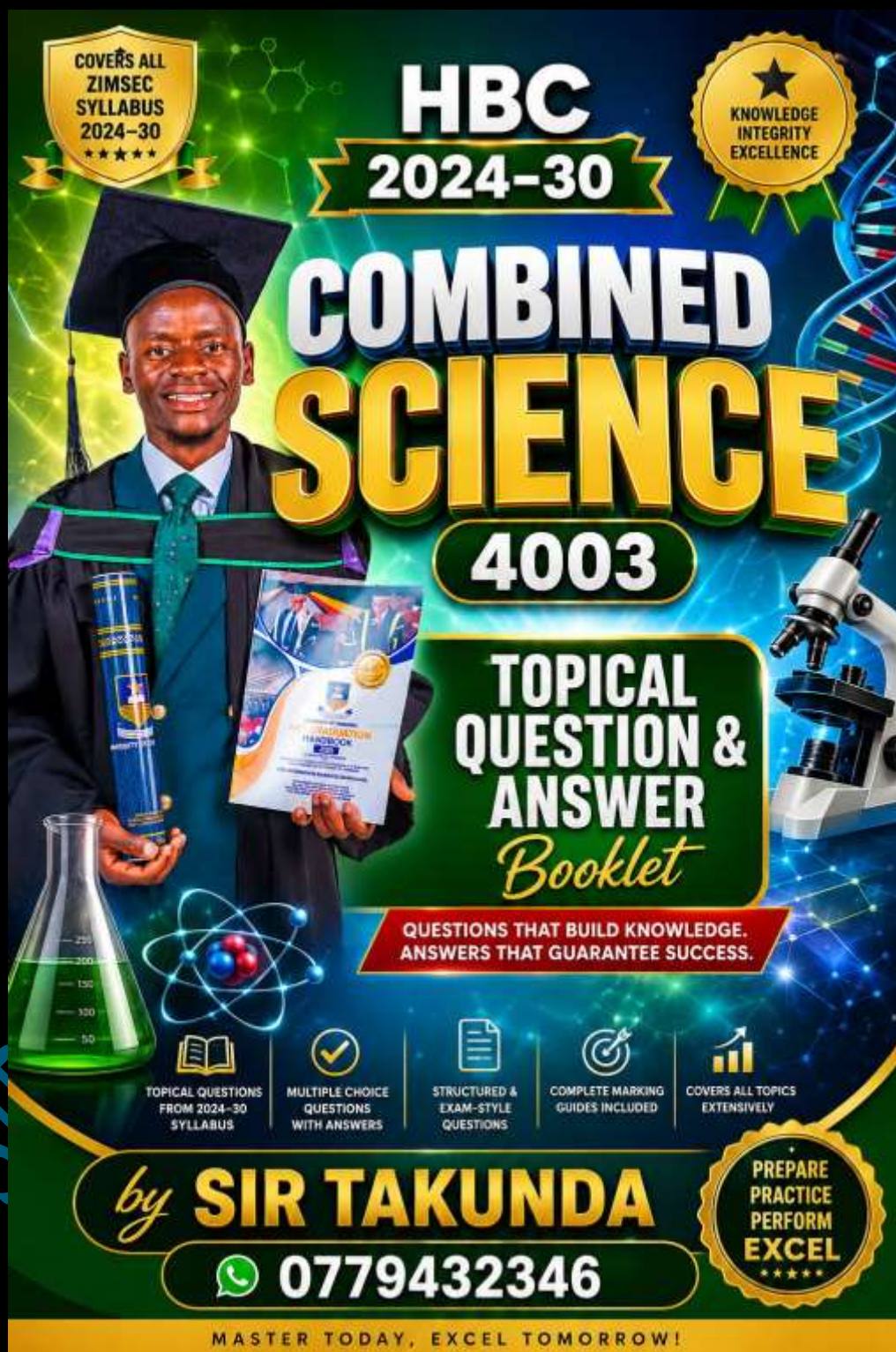




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0779432346

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FORM 3 & 4

TOPIC: OXIDATION AND REDUCTION

Multiple Choice Questions

Question 1

Which one of the following best defines oxidation?

- A. Gain of electrons
- B. Loss of oxygen
- C. Gain of oxygen
- D. Gain of hydrogen

Answer: C

Explanation:

Oxidation is the gain of oxygen, loss of hydrogen or loss of electrons.

Question 2

Reduction is the

- A. gain of oxygen
- B. loss of electrons
- C. gain of electrons
- D. loss of hydrogen

Answer: C

Explanation:

Reduction is the gain of electrons, loss of oxygen or gain of hydrogen.

Question 3

Which memory aid helps learners remember oxidation and reduction in terms of electrons?

- A. LIME
- B. OIL RIG
- C. SMART
- D. KOH

Answer: B

Explanation:

OIL RIG means:

Oxidation Is Loss (of electrons)

Reduction Is Gain (of electrons).

Question 4

Which substance is the main source of iron in the blast furnace?

- A. Coke
- B. Limestone
- C. Haematite
- D. Carbon monoxide

Answer: C

Explanation:

Haematite (Fe_2O_3) is the iron ore used to obtain iron.

Question 5

The main fuel used in the blast furnace is

- A. limestone
- B. charcoal
- C. coke
- D. sulphur

Answer: C

Explanation:

Coke burns to provide heat and produces carbon monoxide for reduction.

Question 6

Which raw material removes impurities from iron ore?

- A. Carbon monoxide
- B. Coke
- C. Limestone
- D. Iron ore

Answer: C

Explanation:

Limestone decomposes to produce calcium oxide which removes silica impurities.

Question 7

Carbon monoxide acts as a

- A. catalyst
- B. reducing agent
- C. oxidising agent

D. solvent

Answer: B

Explanation:

Carbon monoxide removes oxygen from iron oxide and is therefore a reducing agent.

Question 8

Which gas reduces iron(III) oxide inside the blast furnace?

A. Oxygen

B. Carbon dioxide

C. Carbon monoxide

D. Nitrogen

Answer: C

Explanation:

Carbon monoxide reacts with iron(III) oxide to produce iron.

Question 9

Which equation represents the reduction of iron(III) oxide?

A. $C + O_2 \rightarrow CO_2$

B. $CO_2 + C \rightarrow 2CO$

C. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: C

Explanation:

This equation shows iron oxide losing oxygen to form iron.

Question 10

Which substance floats on top of molten iron inside the blast furnace?

- A. Carbon monoxide
- B. Slag
- C. Coke
- D. Limestone

Answer: B

Explanation:

Slag is less dense than molten iron, so it floats and can be removed separately.

Question 11

Which reaction produces carbon monoxide in the blast furnace?

- A. $C + O_2 \rightarrow CO_2$
- B. $CO_2 + C \rightarrow 2CO$
- C. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
- D. $CaO + SiO_2 \rightarrow CaSiO_3$

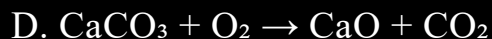
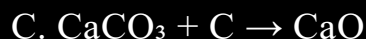
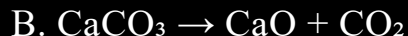
Answer: B

Explanation: Carbon dioxide reacts with hot coke to produce carbon monoxide, the reducing agent.

Question 12

Limestone decomposes according to the equation:

- A. $CaO + CO_2 \rightarrow CaCO_3$



Answer: B

Explanation: Heating limestone produces calcium oxide and carbon dioxide.

Question 13

The main function of calcium oxide in the blast furnace is to

- A. produce oxygen
- B. reduce iron ore
- C. remove silica impurities
- D. cool the furnace

Answer: C

Explanation: Calcium oxide reacts with silica to form slag.

Question 14

Slag is mainly composed of

- A. Iron
- B. Carbon
- C. Calcium silicate
- D. Iron oxide

Answer: C

Explanation: Calcium oxide reacts with silica to produce calcium silicate (slag)

Question 15

Which statement correctly describes slag?

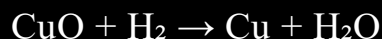
- A. It is heavier than molten iron.
- B. It sinks below molten iron.
- C. It floats on molten iron.
- D. It is pure iron.

Answer: C

Explanation: Slag is less dense than molten iron and floats on top.

Question 16

In the reaction



Copper(II) oxide is

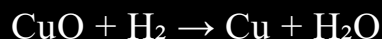
- A. oxidised
- B. reduced
- C. burnt
- D. neutralised

Answer: B

Explanation: Copper(II) oxide loses oxygen to form copper.

Question 17

In the reaction



Hydrogen is

- A. reduced
- B. oxidised
- C. neutralised
- D. decomposed

Answer: B

Explanation: Hydrogen gains oxygen to form water.

Question 18

The black solid in the reduction of copper(II) oxide changes into

- A. white powder
- B. blue crystals
- C. reddish-brown copper
- D. yellow sulphur

Answer: C

Explanation: Copper metal is reddish-brown.

Question 19

Water droplets form during the reduction of copper(II) oxide because

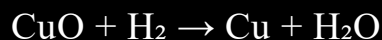
- A. copper melts
- B. oxygen combines with hydrogen
- C. hydrogen evaporates
- D. carbon dioxide condenses

Answer: B

Explanation: Hydrogen reacts with oxygen from copper(II) oxide to form water.

Question 20

The reaction



is classified as

- A. neutralisation
- B. decomposition
- C. redox
- D. precipitation

Answer: C

Explanation: Copper(II) oxide is reduced while hydrogen is oxidised, so oxidation and reduction occur simultaneously.

Question 21

Rusting of iron requires

- A. oxygen only
- B. water only
- C. oxygen and water
- D. carbon dioxide only

Answer: C

Explanation: Iron rusts only when both oxygen and water are present.

Question 22

Which test tube in the rusting experiment will rust?

- A. Nail in dry air only
- B. Nail in boiled water covered with oil
- C. Nail in water exposed to air
- D. Nail completely covered with grease

Answer: C

Explanation: The nail has access to both oxygen and water.

Question 23

The reddish-brown substance formed when iron corrodes is called

- A. slag
- B. soot
- C. rust
- D. scale

Answer: C

Explanation: Rust is hydrated iron(III) oxide formed during corrosion.

Question 24

Which one of the following is **NOT** a method of preventing rusting?

- A. Painting
- B. Greasing

C. Galvanising

D. Heating iron strongly

Answer: D

Explanation: Heating iron does not prevent rusting.

Question 25

Galvanising protects iron by coating it with

A. copper

B. zinc

C. chromium

D. silver

Answer: B

Explanation: Zinc forms a protective coating and also protects iron if scratched.

Question 26

Which method protects iron by covering it with another metal using electrolysis?

A. Painting

B. Greasing

C. Electroplating

D. Plastic coating

Answer: C

Explanation: Electroplating deposits a thin protective metal layer.

Question 27

Which alloy of iron contains chromium?

- A. Mild steel
- B. Cast iron
- C. Stainless steel
- D. Wrought iron

Answer: C

Explanation: Chromium makes stainless steel resistant to corrosion.

Question 28

Which alloy contains the highest percentage of carbon?

- A. Mild steel
- B. Stainless steel
- C. Cast iron
- D. Pure iron

Answer: C

Explanation: Cast iron contains about 2–4% carbon.

Question 29

Which alloy is commonly used to make cooking pots?

- A. Mild steel
- B. Stainless steel
- C. Cast iron
- D. Aluminium

Answer: C

Explanation: Cast iron retains heat well, making it suitable for cooking pots

Question 30

Which alloy is most suitable for making kitchen utensils because it resists rusting?

- A. Cast iron
- B. Mild steel
- C. Stainless steel
- D. Iron ore

Answer: C

Explanation: Stainless steel contains chromium, which protects it from corrosion.

Question 31

Mild steel contains approximately

- A. 0.2–0.8% carbon
- B. 2–4% carbon
- C. 10–20% chromium
- D. 8–10% nickel

Answer: A

Explanation: Mild steel contains a small amount of carbon (about 0.2–0.8%), making it strong and tough.

Question 32

Which alloy is most suitable for building bridges and building frames?

- A. Cast iron

- B. Stainless steel
- C. Mild steel
- D. Pure iron

Answer: C

Explanation: Mild steel is strong, tough, and easy to shape, making it suitable for construction.

Question 33

The main reason stainless steel does not rust easily is because it contains

- A. carbon
- B. zinc
- C. chromium
- D. copper

Answer: C

Explanation: Chromium forms a protective oxide layer that prevents corrosion.

Question 34

Cast iron is preferred for making cooking pots because it

- A. is very soft
- B. conducts electricity well
- C. retains heat for a long time
- D. is lighter than aluminium

Answer: C

Explanation: Cast iron has excellent heat retention, making it ideal for cookware

Question 35

Which property makes cast iron unsuitable for making car bodies?

- A. It is too soft.
- B. It rusts immediately.
- C. It is brittle.
- D. It melts easily.

Answer: C

Explanation: Cast iron is brittle and can crack under impact.

Question 36

Which of the following is an everyday example of oxidation?

- A. Electroplating
- B. Rusting of iron
- C. Electrolysis of water
- D. Condensation

Answer: B

Explanation: Rusting is the oxidation of iron in the presence of oxygen and water

Question 37

Which process uses reduction to produce iron from iron ore?

- A. Rusting
- B. Electroplating

C. Blast furnace extraction

D. Combustion

Answer: C

Explanation: Iron(III) oxide is reduced to iron in the blast furnace.

Question 38

Which process is an example of a redox reaction occurring in living organisms?

A. Digestion

B. Respiration

C. Evaporation

D. Filtration

Answer: B

Explanation: Respiration involves oxidation of glucose and reduction of oxygen to release energy.

Question 39

A reducing agent is a substance that

A. gains oxygen

B. removes oxygen from another substance

C. accepts electrons

D. causes corrosion

Answer: B

Explanation: A reducing agent removes oxygen or donates electrons to another substance.

Question 40

An oxidising agent is a substance that

- A. loses oxygen
- B. donates electrons
- C. supplies oxygen or accepts electrons
- D. prevents rusting

Answer: C

Explanation: An oxidising agent causes oxidation by supplying oxygen or accepting electrons.

Question 41

Which process in the blast furnace is a reduction reaction?

- A. $C + O_2 \rightarrow CO_2$
- B. $CO_2 + C \rightarrow 2CO$
- C. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
- D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: C

Explanation: Iron(III) oxide loses oxygen to form iron, making it a reduction reaction.

Question 42

Which raw material acts as both a fuel and a reducing agent in the blast furnace?

- A. Iron ore
- B. Limestone

C. Coke

D. Slag

Answer: C

Explanation: Coke burns to produce heat and forms carbon monoxide, which reduces iron ore.

Question 43

Why is limestone added to the blast furnace?

A. To increase the temperature

B. To produce oxygen

C. To remove silica impurities by forming slag

D. To reduce iron oxide

Answer: C

Explanation: Limestone decomposes to calcium oxide, which reacts with silica impurities to form slag.

Question 44

Which statement about rust is correct?

A. Rust protects iron from further corrosion.

B. Rust is stronger than iron.

C. Rust is porous and allows corrosion to continue.

D. Rust is pure iron.

Answer: C

Explanation: Rust flakes off easily, exposing fresh iron to further corrosion.

Question 45

Which of the following is **NOT** an alloy of iron?

- A. Mild steel
- B. Stainless steel
- C. Cast iron
- D. Brass

Answer: D

Explanation: Brass is an alloy of copper and zinc, not iron.

Question 46

Which alloy would be the best choice for making surgical instruments?

- A. Mild steel
- B. Cast iron
- C. Stainless steel
- D. Pure iron

Answer: C

Explanation: Stainless steel is corrosion-resistant, strong and hygienic, making it ideal for medical instruments.

Question 47

Which process is an example of oxidation occurring in everyday life?

- A. Electroplating a spoon
- B. Reduction of copper oxide
- C. Burning charcoal

D. Formation of iron from iron ore

Answer: C

Explanation: Burning charcoal is combustion, which is an oxidation reaction because carbon gains oxygen.

Question 48

During electroplating, metal ions gain electrons at the

- A. anode
- B. cathode
- C. electrolyte
- D. battery

Answer: B

Explanation: Reduction occurs at the cathode where metal ions gain electrons and are deposited as metal.

Question 49

Which one of the following is an example of a reducing agent?

- A. Oxygen
- B. Carbon monoxide
- C. Water
- D. Chlorine

Answer: B

Explanation: Carbon monoxide removes oxygen from iron(III) oxide and is therefore a reducing agent.

Question 50

Which statement correctly describes a redox reaction?

- A. Only oxidation occurs.
- B. Only reduction occurs.
- C. Oxidation and reduction occur at the same time.
- D. No electrons are transferred.

Answer: C

Explanation: In every redox reaction, oxidation and reduction always occur simultaneously because electrons lost by one substance are gained by another.

Answer Summary

Qn Ans Qn Ans Qn Ans Qn Ans Qn Ans

1	C	11	B	21	C	31	A	41	C
2	C	12	B	22	C	32	C	42	C
3	B	13	C	23	C	33	C	43	C
4	C	14	C	24	D	34	C	44	C
5	C	15	C	25	B	35	C	45	D
6	C	16	B	26	C	36	B	46	C
7	B	17	B	27	C	37	C	47	C
8	C	18	C	28	C	38	B	48	B
9	C	19	B	29	C	39	B	49	B
10	B	20	C	30	C	40	C	50	C

TOPIC: OXIDATION AND REDUCTION

50 Structured Questions

Question 1 (4 Marks)

Define the following terms:

- a) Oxidation (2)

b) Reduction (2)

Marking Guide

a)

- Gain of oxygen (1)
- OR loss of hydrogen OR loss of electrons (1)

b)

- Loss of oxygen (1)
- OR gain of hydrogen OR gain of electrons (1)

Total: 4 Marks

Question 2 (6 Marks)

State three ways in which oxidation can occur and three ways in which reduction can occur.

Marking Guide

Oxidation

- Gain of oxygen (1)
- Loss of hydrogen (1)
- Loss of electrons (1)

Reduction

- Loss of oxygen (1)
- Gain of hydrogen (1)
- Gain of electrons (1)

Total: 6 Marks

Question 3 (6 Marks)

State the raw materials used in the blast furnace for extracting iron and give one function of each.

Marking Guide

Raw Material	Function
Iron ore (1)	Source of iron (1)
Coke (1)	Fuel/reducing agent (1)
Limestone (1)	Removes impurities/forms slag (1)

Total: 6 Marks**Question 4 (8 Marks)**

Describe the production of iron in the blast furnace.

Marking Guide

Award one mark for each point.

- Iron ore added (1)
- Coke added (1)
- Limestone added (1)
- Hot air blown into furnace (1)
- Coke burns producing carbon dioxide (1)
- Carbon monoxide formed (1)
- Carbon monoxide reduces iron oxide (1)
- Molten iron collected at bottom (1)

Total: 8 Marks**Question 5 (6 Marks)**

State the functions of the following in the blast furnace.

- a) Coke (2)
- b) Limestone (2)
- c) Hot air (2)

Marking Guide

- a)

- Fuel (1)
- Produces carbon monoxide/reducing agent (1)

b)

- Produces calcium oxide (1)
- Removes silica impurities/forms slag (1)

c)

- Supplies oxygen (1)
- Supports combustion (1)

Total: 6 Marks

Question 6 (8 Marks)

The following reactions occur inside a blast furnace.

- Write the equation for the formation of carbon dioxide. (2)
- Write the equation for the formation of carbon monoxide. (2)
- Write the equation for the reduction of iron(III) oxide. (2)
- Write the equation for slag formation. (2)

Marking Guide

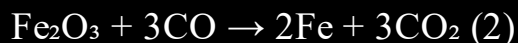
a)



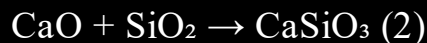
b)



c)



d)



Total: 8 Marks

Question 7 (6 Marks)

Explain why slag floats on molten iron in the blast furnace.

Marking Guide

- Slag is calcium silicate (1)
- Slag has lower density (2)
- Molten iron has higher density (2)
- Therefore slag floats on top (1)

Total: 6 Marks

Question 8 (6 Marks)

Copper(II) oxide is heated in a stream of hydrogen gas.

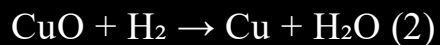
- a) State two observations. (2)
- b) Write the equation. (2)
- c) State which substance is reduced. (2)

Marking Guide

a)

- Black solid changes to reddish-brown copper (1)
- Water droplets form (1)

b)



c)

Copper(II) oxide is reduced (2)

Total: 6 Marks

Question 9 (8 Marks)

Explain why the reaction between copper(II) oxide and hydrogen is called a redox reaction.

Marking Guide

- Copper oxide loses oxygen (2)
- Copper oxide is reduced (2)
- Hydrogen gains oxygen (2)
- Hydrogen is oxidised (2)

Total: 8 Marks

Question 10 (6 Marks)

Complete the table below.

Process	Example
Oxidation _____	
Reduction _____	

Give one example of each.

Marking Guide

Oxidation

Example:

- Rusting of iron (1)
- Burning magnesium (1)
- Burning charcoal (1)

Reduction

Example:

- $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$ (1)

- $\text{Fe}_2\text{O}_3 \rightarrow \text{Fe}$ (1)
- Electroplating (reduction at cathode) (1)

Award any two correct examples.

Total: 6 Marks

Question 11 (6 Marks)

State the conditions necessary for rusting of iron.

Marking Guide

- Presence of oxygen (2)
- Presence of water/moisture (2)
- Both oxygen and water must be present together (2)

Total: 6 Marks

Question 12 (8 Marks)

Explain how rusting occurs.

Marking Guide

Award one mark for each point.

- Iron reacts with oxygen (1)
- Water is present (1)
- Iron is oxidised (1)
- Hydrated iron(III) oxide forms (1)
- Rust is reddish-brown (1)
- Rust is porous/flaky (1)
- Fresh iron becomes exposed (1)
- Rusting continues (1)

Total: 8 Marks

Question 13 (6 Marks)

State six methods used to prevent rusting.

Marking Guide

Any six:

- Painting (1)
- Greasing (1)
- Oiling (1)
- Galvanising (1)
- Electroplating (1)
- Plastic coating (1)
- Using stainless steel (1)

Award 1 mark each (maximum 6).

Total: 6 Marks

Question 14 (8 Marks)

Explain how each of the following methods prevents rusting.

- a) Painting (2)
- b) Galvanising (2)
- c) Greasing (2)
- d) Electroplating (2)

Marking Guide

- a)
 - Paint prevents oxygen and water reaching iron (2)
- b)
 - Zinc coating protects iron from corrosion (2)
- c)

- Grease forms a waterproof barrier (2)

d)

- Protective metal coating prevents contact with air and water (2)

Total: 8 Marks

Question 15 (6 Marks)

State three harmful effects of rusting.

Marking Guide

Any three:

- Weakens bridges (2)
- Weakens buildings (2)
- Damages vehicles (2)
- Damages machinery (2)
- Causes water pipe leaks (2)
- Leads to accidents (2)

Award 2 marks for each correct effect (maximum 6).

Total: 6 Marks

Question 16 (8 Marks)

A learner carried out an investigation using three test tubes A, B and C containing iron nails.

State:

- a) Which test tube rusted? (2)
- b) Why did it rust? (2)
- c) Why did Test Tube B not rust? (2)
- d) Why did Test Tube C not rust? (2)

Marking Guide

a)

- Test Tube A (2)

b)

- Oxygen and water were both present (2)

c)

- Oxygen absent (boiled water covered with oil) (2)

d)

- Water absent (dry air only) (2)

Total: 8 Marks**Question 17 (6 Marks)**

Draw a conclusion from the rusting experiment.

Marking Guide

- Iron rusts only when oxygen is present (2)
- Iron rusts only when water is present (2)
- Both oxygen and water are needed together (2)

Total: 6 Marks**Question 18 (8 Marks)**

State the independent, dependent and two controlled variables in the rusting investigation.

Marking Guide

Independent variable

- Presence/absence of oxygen and water (2)

Dependent variable

- Amount of rust formed (2)

Controlled variables (any two)

- Size of nail (1)
- Type of nail (1)
- Temperature (1)
- Time (1)
- Size of test tube (1)

Award 2 marks.

Total: 8 Marks

Question 19 (6 Marks)

State three ways in which the rusting investigation could be improved.

Marking Guide

Any three:

- Use identical nails (2)
- Use airtight test tubes (2)
- Measure amount of rust (2)
- Keep temperature constant (2)
- Observe for the same length of time (2)

Award 2 marks each (maximum 6).

Total: 6 Marks

Question 20 (8 Marks)

Explain why stainless steel does not rust easily.

Marking Guide

- Stainless steel contains chromium (2)
- Chromium reacts with oxygen (2)

- Protective chromium oxide layer forms (2)
- Prevents oxygen and water reaching iron (2)

Total: 8 Marks

Question 21 (6 Marks)

Name the three common alloys of iron.

Marking Guide

- Mild steel (2)
- Stainless steel (2)
- Cast iron (2)

Total: 6 Marks

Question 22 (6 Marks)

Define an alloy.

Marking Guide

- An alloy is a mixture of a metal (2)
- With one or more other elements (2)
- To improve its properties (2)

Total: 6 Marks

Question 23 (8 Marks)

State the approximate percentage composition of the following alloys:

- a) Mild steel (2)
- b) Stainless steel (3)
- c) Cast iron (3)

Marking Guide

- a)

- Iron 99.2–99.8% (1)
- Carbon 0.2–0.8% (1)

b)

- Iron about 70–75% (1)
- Chromium 10–20% (1)
- Nickel 8–10% (1)

c)

- Iron 96–98% (1)
- Carbon 2–4% (1)
- Any correct indication that cast iron contains the highest carbon content (1)

Total: 8 Marks

Question 24 (6 Marks)

State three properties of mild steel.

Marking Guide

Any three:

- Strong (2)
- Tough (2)
- Ductile (2)
- Malleable (2)
- Easy to weld (2)
- Easy to shape (2)

Award 2 marks each (maximum 6).

Total: 6 Marks

Question 25 (6 Marks)

Explain why mild steel is widely used in construction.

Marking Guide

Any three:

- Strong (2)
- Tough (2)
- Easy to shape (2)
- Easy to weld (2)
- Relatively inexpensive (2)
- Suitable for heavy structures (2)

Award 2 marks each (maximum 6).

Total: 6 Marks

Question 26 (8 Marks)

State four properties of stainless steel and explain why it is suitable for making kitchen utensils.

Marking Guide

Properties (any four)

- Corrosion resistant (1)
- Strong (1)
- Hard (1)
- Durable (1)

Explanation

- Does not rust (2)
- Safe for food preparation/easy to clean (2)

Total: 8 Marks

Question 27 (6 Marks)

State three uses of stainless steel.

Marking Guide

Any three:

- Kitchen utensils (2)
- Surgical instruments (2)
- Kitchen sinks (2)
- Water tanks (2)
- Food-processing equipment (2)
- Hospital equipment (2)

Award 2 marks each (maximum 6).

Total: 6 Marks

Question 28 (8 Marks)

State four properties of cast iron.

Marking Guide

Any four:

- Hard (2)
- Brittle (2)
- Retains heat well (2)
- Strong under compression (2)
- Resistant to wear (2)

Award 2 marks each (maximum 8).

Total: 8 Marks

Question 29 (6 Marks)

Explain why cast iron is used to make cooking pots.

Marking Guide

Any three:

- Retains heat well (2)
- Strong (2)
- Durable (2)
- Withstands high temperatures (2)

Award 2 marks each (maximum 6).

Total: 6 Marks

Question 30 (8 Marks)

Complete the table below.

Alloy	One Property	One Use
Mild Steel	_____	_____
Stainless Steel	_____	_____
Cast Iron	_____	_____

Marking Guide

Alloy	Property	Use
Mild Steel	Strong/tough (1)	Construction/bridges/vehicle bodies (1)
Stainless Steel	Corrosion resistant (1)	Kitchen utensils/surgical instruments (1)
Cast Iron	Hard and brittle/retains heat (1)	Cooking pots/engine blocks/drain covers (1)

Award 1 mark for each correct entry.

Total: 8 Marks

Question 31 (8 Marks)

Define the following terms and give one example of each.

a) Oxidising agent (4)

b) Reducing agent (4)

Marking Guide

a) Oxidising agent

- Substance that causes oxidation (1)
- Accepts electrons or supplies oxygen (2)
- Example: Oxygen/iron(III) oxide (1)

b) Reducing agent

- Substance that causes reduction (1)
- Donates electrons or removes oxygen (2)
- Example: Carbon monoxide/hydrogen (1)

Total: 8 Marks**Question 32 (6 Marks)**

State whether each of the following substances acts as an oxidising agent or a reducing agent.

a) Carbon monoxide

b) Hydrogen

c) Oxygen

Marking Guide

- Carbon monoxide – Reducing agent (2)
- Hydrogen – Reducing agent (2)
- Oxygen – Oxidising agent (2)

Total: 6 Marks**Question 33 (8 Marks)**

The equation below represents a redox reaction.



Answer the following questions.

- a) Which substance is reduced? (2)
- b) Which substance is oxidised? (2)
- c) Which substance is the reducing agent? (2)
- d) Which substance is the oxidising agent? (2)

Marking Guide

- a) Iron(III) oxide (2)
- b) Carbon monoxide (2)
- c) Carbon monoxide (2)
- d) Iron(III) oxide (2)

Total: 8 Marks

Question 34 (6 Marks)

Explain why carbon monoxide is called a reducing agent.

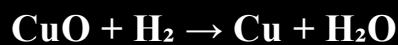
Marking Guide

- Removes oxygen from iron oxide (2)
- Causes iron oxide to be reduced (2)
- Is itself oxidised to carbon dioxide (2)

Total: 6 Marks

Question 35 (8 Marks)

Study the reaction below.



Answer the following questions.

- a) Which substance gains oxygen? (2)
- b) Which substance loses oxygen? (2)
- c) Which substance is oxidised? (2)
- d) Which substance is reduced? (2)

Marking Guide

- a) Hydrogen (2)
- b) Copper(II) oxide (2)
- c) Hydrogen (2)
- d) Copper(II) oxide (2)

Total: 8 Marks

Question 36 (6 Marks)

State three examples of oxidation reactions in everyday life.

Marking Guide

Any three:

- Rusting of iron (2)
- Burning charcoal (2)
- Burning magnesium (2)
- Respiration (2)
- Burning fuel (2)

Award **2 marks each** (maximum 6).

Total: 6 Marks

Question 37 (8 Marks)

State four examples of reduction reactions used in industry.

Marking Guide

Any four:

- Extraction of iron from iron ore (2)
- Reduction of copper oxide by hydrogen (2)
- Electroplating (reduction at cathode) (2)
- Electrolysis of molten lead bromide (lead ions gain electrons) (2)
- Extraction of metals from ores (2)

Award **2 marks each** (maximum 8).

Total: 8 Marks

Question 38 (6 Marks)

Explain why respiration is considered a redox reaction.

Marking Guide

- Glucose is oxidised (2)
- Oxygen is reduced (2)
- Both oxidation and reduction occur simultaneously (2)

Total: 6 Marks

Question 39 (8 Marks)

Differentiate between oxidation and reduction using electrons and oxygen.

Marking Guide

Oxidation

Loss of electrons (2)

Gain of oxygen (2)

Reduction

Gain of electrons (2)

Loss of oxygen (2)

Total: 8 Marks

Question 40 (6 Marks)

A learner says:

"Whenever oxidation occurs, reduction must also occur."

Explain whether this statement is correct.

Marking Guide

- Statement is correct (2)
- Oxidation involves loss of electrons (2)
- Electrons lost must be gained by another substance (reduction) (2)

Total: 6 Marks

Question 41 (8 Marks)

A learner visits a steel manufacturing plant and observes a blast furnace.

Answer the following questions.

- a) Name the furnace used to extract iron. (2)
- b) State the main ore used. (2)
- c) Name the reducing agent. (2)
- d) State one product other than iron. (2)

Marking Guide

- a) Blast furnace (2)

- b) Haematite (Fe_2O_3) (2)
- c) Carbon monoxide (2)
- d) Slag/Carbon dioxide (2)

Total: 8 Marks

Question 42 (8 Marks)

Explain why carbon monoxide is preferred to carbon in reducing iron(III) oxide inside the blast furnace.

Marking Guide

Award one mark for each point.

- Carbon monoxide is a gas (2)
- It moves easily through the furnace (2)
- It comes into contact with iron ore more effectively (2)
- It removes oxygen from iron oxide efficiently (2)

Total: 8 Marks

Question 43 (8 Marks)

State four industrial applications of oxidation and reduction.

Marking Guide

Any four:

- Extraction of iron (2)
- Electroplating (2)
- Manufacture of steel (2)
- Respiration (2)
- Combustion of fuels (2)
- Battery reactions (2)

Award **2 marks each** (maximum 8).

Total: 8 Marks

Question 44 (8 Marks)

A factory produces iron cooking pots.

Explain why cast iron is selected instead of mild steel.

Marking Guide

Any four:

- Retains heat well (2)
- Hard (2)
- Strong under compression (2)
- Durable (2)

Award **2 marks each** (maximum 8).

Total: 8 Marks

Question 45 (8 Marks)

State four advantages of stainless steel over ordinary iron.

Marking Guide

Any four:

- Corrosion resistant (2)
- Attractive appearance (2)
- Strong (2)
- Hygienic/easy to clean (2)
- Long lasting (2)

Award **2 marks each** (maximum 8).

Total: 8 Marks

Question 46 (8 Marks)

A learner leaves an iron nail outside during the rainy season.

Explain why the nail rusts faster than one kept indoors.

Marking Guide

Award one mark for each point.

- Rain provides water (2)
- Air supplies oxygen (2)
- Moist conditions speed up rusting (2)
- Both oxygen and water are always available outdoors (2)

Total: 8 Marks

Question 47 (8 Marks)

Suggest four ways a farmer can prevent farm tools from rusting.

Marking Guide

Any four:

- Paint them (2)
- Oil them (2)
- Grease them (2)
- Store in a dry place (2)
- Galvanise them (2)
- Electroplate them (2)

Award 2 marks each (maximum 8).

Total: 8 Marks

Question 48 (8 Marks)

A learner heats copper(II) oxide in hydrogen gas.

State:

- a) The colour of copper(II) oxide before heating. (2)
- b) The colour of copper formed. (2)
- c) The colourless liquid produced. (2)

d) The type of reaction. (2)

Marking Guide

a) Black (2)

b) Reddish-brown (2)

c) Water (2)

d) Redox reaction (2)

Total: 8 Marks

Question 49 (8 Marks)

State four observations made during the reduction of copper(II) oxide by hydrogen.

Marking Guide

Any four:

- Black copper oxide changes colour (2)
- Reddish-brown copper forms (2)
- Water droplets appear (2)
- Hydrogen passes through the tube (2)
- Heat is supplied (2)

Award **2 marks each** (maximum 8).

Total: 8 Marks

Question 50 (10 Marks)

Oxidation and reduction play an important role in industry.

Discuss this statement using suitable examples.

Marking Guide

Award **any five well-explained points (2 marks each)**.

Possible answers:

- Extraction of iron from iron ore (2)
- Electroplating metals (2)
- Manufacture of steel (2)
- Rusting and corrosion (2)
- Combustion of fuels (2)
- Respiration (2)
- Battery reactions (2)
- Reduction of copper oxide (2)
- Manufacture of industrial metals (2)

Maximum: 10 Marks